Intro to Machine Learning
Practice Problems 1: Bias, Variance and the Curse of Dimensionality

Practice problems, not turned in — Solutions

Directions: These practice problems go with Week 1 (lecture: *What Is Machine Learning? Bias, Variance, and the Curse of Dimensionality*). They are not turned in, but the quizzes draw on them, so work through each one before looking at the solutions. Each problem is adapted from the exercise cited next to it; see the book for the original wording.

1. **Flexible or inflexible?** (ISL 2.4, Exercise 1; ISL 2.1.3, 2.2.2.) For each situation, say whether you would generally expect a *flexible* learning method to perform better or worse than an *inflexible* one, and justify your answer using bias and variance.

   (a) The sample size n is very large and the number of predictors p is small.

   (b) The number of predictors p is very large and n is small.

   (c) The relationship between the predictors and the response is highly non-linear.

   (d) The error variance $\sigma^2 = \mathrm{Var}(\varepsilon)$ is very large.

   Solution.

   (a) **Flexible is better.** The variance of a fitted method shrinks as n grows (for KNN it is σ^2/K with K allowed to grow with n; for least squares it is about $(p+1)\sigma^2/n$). With lots of data, a flexible method can follow the true f closely, cutting bias, without paying much in variance.

   (b) **Inflexible is better.** With few observations a flexible method has high variance and will fit the noise. With many predictors it also meets the curse of dimensionality: neighbourhoods stop being local, so local methods lose their low-bias advantage. An inflexible method borrows strength from all n points through its assumptions, and its variance grows only like p/n (lecture §5.5).

   (c) **Flexible is better.** An inflexible method, such as a linear fit, cannot represent a highly non-linear f. Its bias stays large however much data you have, and a flexible method removes most of that bias.

   (d) **Inflexible is better.** The irreducible error σ^2 is the same for every method, so it does not decide the comparison. What matters is that the *variance* term scales with σ^2 (σ^2/K for KNN, $(p+1)\sigma^2/n$ for least squares). Noisy responses make a flexible fit chase the noise, and its variance grows faster than an inflexible fit's.

2. **Regression or classification? Inference or prediction?** (ISL 2.4, Exercise 2; ISL 2.1.1, 2.1.5.) For each scenario, say whether it is a regression or a classification problem, whether the main interest is inference or prediction, and give n and p.

   (a) For each of the 500 largest US firms we record profit, number of employees, industry and CEO salary, and we want to understand which factors affect CEO salary.

   (b) Before launching a new product, we want to know whether it will succeed or fail. We have 20 similar past products, and for each we know whether it succeeded, its price, its marketing budget, the competitor's price and ten other variables.

   (c) We want to predict the weekly % change in the USD/Euro exchange rate from the weekly % changes in the US, British and German stock markets, using weekly data for all of 2012.

   Solution.

   (a) **Regression** (CEO salary is quantitative) and **inference** (we want to know which factors affect salary). $n = 500$ firms and $p = 3$ predictors: profit, number of employees and industry.

   (b) **Classification** (success or failure) and **prediction** (we want to know how the new product will do, not why). $n = 20$ products and $p = 13$ predictors: price, marketing budget, competition price and ten others.

   (c) **Regression** (% change is quantitative) and **prediction**. $n = 52$ weeks in 2012 and $p = 3$ predictors: the % changes in the US, British and German markets.

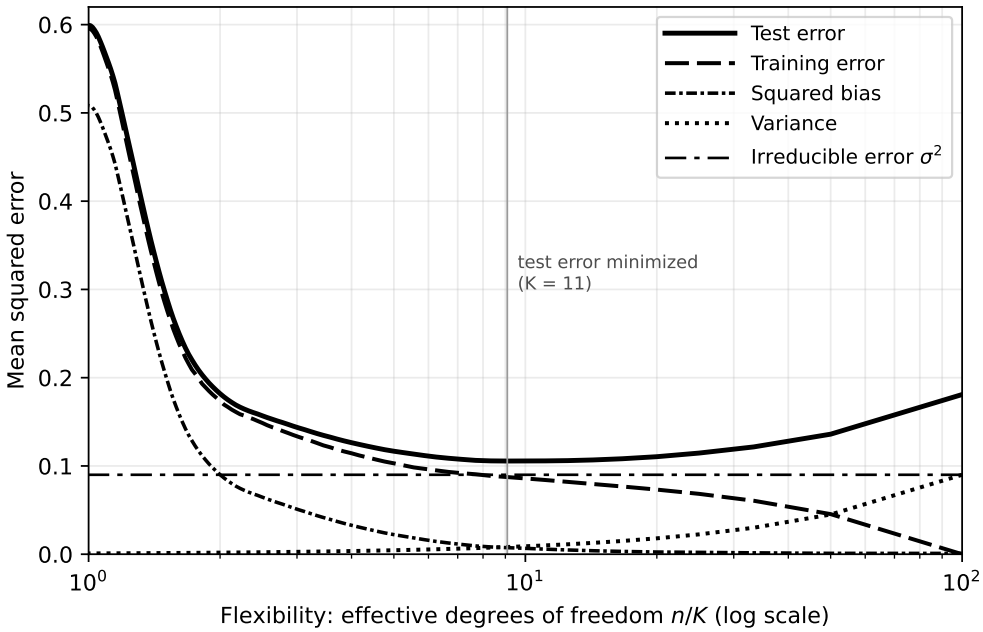
3. **The bias–variance picture.** (ISL 2.4, Exercise 3; ISL 2.2.2; lecture §4.)

   (a) On a single plot, sketch typical curves for squared bias, variance, training error, test error and irreducible (Bayes) error as a method goes from less flexible to more flexible. Put flexibility on the x-axis and label all five curves.

   (b) Explain why each curve has the shape you drew.

   Solution.

   (a) The figure below shows the five curves. Rather than a freehand sketch, it plots the exact values for KNN regression from lecture §4.3: $f(x) = \sin(2\pi x)$, $\sigma^2 = 0.09$, $n = 100$, with flexibility measured by the effective degrees of freedom n/K, so that $K = 100$ is on the left and $K = 1$ on the right.



Squared bias, variance, irreducible error, test error and training error for KNN regression as flexibility increases (lecture §4.3). Test error is minimized at $K = 11$.

(b)
- **Squared bias** decreases. More flexible methods can represent more shapes, so the *average* fit gets closer to the true f. For KNN, smaller K averages over neighbours nearer to x_0, where f is closer to $f(x_0)$.
- **Variance** increases. A more flexible fit depends on fewer data points, or bends to follow individual points, so it changes more from one training set to the next. For KNN it is exactly σ^2/K.
- **Irreducible error** is flat at σ^2. It comes from ε, which no method can predict from X, so it doesn't depend on the method.
- **Test error** is U-shaped. By the bias–variance decomposition it equals $\sigma^2 + \text{bias}^2 +$ variance. At first the fall in bias outweighs the rise in variance; past the minimum the reverse holds. It never drops below σ^2.
- **Training error** decreases monotonically and can fall below σ^2, even to 0. The method is judged on the same data it was fitted to, so extra flexibility always helps in-sample. Its gap from test error, the optimism, is $2\sigma^2 \operatorname{tr}(S)/n$ for a linear smoother (lecture §3), and that grows with flexibility.

4. **Flexibility: pros and cons.** (ISL 2.4, Exercise 5; ISL 2.1.3.) What are the advantages and disadvantages of a very flexible approach to regression or classification, compared with a less flexible one? When would you prefer each?

   Solution. **Advantages of a flexible approach:** it can fit a wide range of shapes for f, so it has low bias when the true relationship is complicated or non-linear, and it may give better predictions.

   Disadvantages: it has higher variance and can overfit by following the noise. It usually needs more data, especially as p grows (the curse of dimensionality). It is harder to interpret, and often more expensive to fit.

Prefer a more flexible approach when prediction accuracy is the goal, n is large relative to p, the signal-to-noise ratio is high, and f is believed to be non-linear.

Prefer a less flexible approach when the goal is inference or interpretability, n is small or p is large, the data are noisy, or f is believed to be close to the simpler form, such as linear.

5. **KNN by hand.** (ISL 2.4, Exercise 7; ISL 2.2.3.) The training set below has six observations, three predictors and a qualitative response. We want to predict Y at the test point $X_1 = X_2 = X_3 = 0$ using K-nearest neighbours with Euclidean distance.

Obs.	X_1	X_2	X_3	Y
1	0	3	0	Red
2	2	0	0	Red
3	0	1	3	Red
4	0	1	2	Green
5	-1	0	1	Green
6	1	1	1	Red

(a) Compute the Euclidean distance from each observation to the test point.

(b) What is the prediction with $K = 1$? Why?

(c) What is the prediction with $K = 3$? Why?

(d) If the Bayes decision boundary is highly non-linear, would you expect the best K to be large or small? Why?

Solution.

(a) The distance from (x_1, x_2, x_3) to the origin is $\sqrt{x_1^2 + x_2^2 + x_3^2}$:

Obs.	Squared distance	Distance	Y
1	$0 + 9 + 0 = 9$	3	Red
2	$4 + 0 + 0 = 4$	2	Red
3	$0 + 1 + 9 = 10$	$\sqrt{10} \approx 3.16$	Red
4	$0 + 1 + 4 = 5$	$\sqrt{5} \approx 2.24$	Green
5	$1 + 0 + 1 = 2$	$\sqrt{2} \approx 1.41$	Green
6	$1 + 1 + 1 = 3$	$\sqrt{3} \approx 1.73$	Red

(b) **Green.** The single nearest neighbour is observation 5 (distance $\sqrt{2} \approx 1.41$), which is Green.

(c) **Red.** The three nearest are observations 5 (Green, 1.41), 6 (Red, 1.73) and 2 (Red, 2). The estimated probabilities are $\widehat{P}(\text{Red}) = 2/3$ and $\widehat{P}(\text{Green}) = 1/3$, so KNN predicts the majority class, Red.

(d) **Small.** A small K is flexible: the decision boundary can bend to follow a highly non-linear Bayes boundary, so its bias is low. A large K averages over a wide neighbourhood and gives a smoother, nearly linear boundary, which is too biased here (ISL Figure 2.16). K should still not be so small that variance dominates; in practice, choose it by cross-validation.

6. **The curse of dimensionality for KNN.** (ISL 4.8, Exercise 4; lecture §5.1.) Observations are uniformly distributed on $[0, 1]$ in each feature. To predict at a test point, we use only training observations within the 10% of each feature's range closest to that test point. For example, with $p = 1$ and test point $X = 0.6$, we use observations with $X \in [0.55, 0.65]$.

   (a) With $p = 1$, on average what fraction of the observations do we use?

   (b) With $p = 2$, and (X_1, X_2) uniform on $[0, 1]^2$, we use observations within 10% of the range of X_1 *and* within 10% of the range of X_2. What fraction do we use?

   (c) What fraction do we use with $p = 100$?

   (d) Use (a)–(c) to argue that when p is large, very few training observations are "near" any test point.

   (e) Now build a p-dimensional hypercube centred on the test point that contains, on average, 10% of the training observations. For $p = 1$, 2 and 100, how long is each side? Comment.

Solution.

(a) **10%.** The window has width 0.1, and X is uniform, so it holds a fraction 0.1 of the observations. Near the ends, take the window to be the 10% of the range closest to the test point, e.g. $[0, 0.1]$ for $X = 0.02$, so the window always has width 0.1.

If you instead cut off a centred window at the boundary, the window holds $x + 0.05$ for $x < 0.05$ (and symmetrically near 1). The average is then

$$0.9 \times 0.1 + 2 \times 0.05 \times \frac{0.05 + 0.1}{2} = 0.09 + 0.0075 = 0.0975,$$

so 9.75%. Either answer is fine if you say which window you used.

(b) **1%.** The two features are independent and uniform, so

$$P(\text{within 10\% in } X_1 \text{ and in } X_2) = 0.1 \times 0.1 = 0.01.$$

(c) $0.1^{100} = 10^{-100}$**.** The same argument multiplies p independent factors of 0.1. That is effectively zero: no training set will ever contain a point in the window.

(d) The fraction of points used is 0.1^p, which shrinks *exponentially* in p. To have even one neighbour on average you need $n \approx 10^p$ observations. So for large p, a neighbourhood that is "local" in every coordinate is empty. To find any neighbours at all, KNN must look far from the test point, and then the neighbours' responses no longer reflect $f(x_0)$, so the bias is large.

(e) A cube of side ℓ in $[0, 1]^p$ has volume ℓ^p, so it contains 10% of uniform data when $\ell^p = 0.1$, that is $\ell = 0.1^{1/p}$:

$$p = 1 : \ \ell = 0.1, \qquad p = 2 : \ \ell = 0.1^{1/2} \approx 0.316, \qquad p = 100 : \ \ell = 0.1^{1/100} \approx 0.977.$$

With $p = 100$, the "neighbourhood" that holds 10% of the data spans 97.7% of the range of *every* feature. It is not local in any sense, and that is why KNN breaks down in high dimensions. This is ESL's $e_p(r) = r^{1/p}$ (ESL 2.5, Figure 2.6).